



SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY
DEPARTMENT
BS Computer Science
COURSE INFORMATION SHEET

Session:	Fall-2024
Course Title:	Compiler Construction
Course Code:	CS-310T
Credit Hours:	3
Semester:	5th
Pre-Requisite:	Theory of Automata and Formal Languages(CS224T)
Instructor Name:	Ms. Mehwish Wahid & Dr. Anam Akbar
Email and Contact Information:	mewahid@ssuet.edu.pk ; anam.akbar@ssuet.edu.pk HS-16
Office Hours:	Tuesdays & Wednesdays (2:00pm to 3:30pm)
Mode of Teaching:	Synchronous/Asynchronous/Hybrid/Blended

COURSE OBJECTIVE:

Provide an understanding of the fundamental principles in Compiler Construction. Learn the process of mapping and design the self-developed language to implement different parsing techniques in order to make efficient parsers. Provide the skills needed for building compilers for various situations that one may encounter in a career in Computer Science.

COURSE OUTLINE:

Introduction to Interpreter and compilers, Compiler Techniques and Methodology. Organization of Compilers. Lexical and Syntax Analysis. Parsing techniques, Types of parser, Type checking, Semantic analyzer, Object code generation and optimization, detection and recovery from errors . Contrast between compilers and interpreter.

COURSE LEARNING OUTCOMES (CLOs):

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom's Taxonomy
1	Discuss the fundamental concepts of the basic structure of compiler and recall R.E and FA.	PLO_1 (Academic Education)	C2 (Understanding)
2	Explain different methods and techniques through which one can design its own Language.	PLO_2 (Knowledge for Computing Problem)	C2 (Understanding)
3	Apply the Different parsers and draw connections of their own generated Grammar in any host Language	PLO_3 (Problem Analysis)	C3 (Applying)



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RELATIONSHIP BETWEEN ASSESSMENT TOOLS AND CLOs:

Assessment Tools	CLO1	CLO2	CLO3
Quizzes	3 (9.6%)	3 (8.3%)	4 (12%)
Assignments	3 (9.6%)	3 (8.3%)	4 (12%)
Mid	15(48.3%)	15(41.6)	-----
Final	10(32.2%)	15(41.6%)	25(76%)
Total	31 (31%)	36 (26%)	33 (33%)

GRADING POLICY:

Assessment Tools	Percentage	Marks
Quizzes	10%	10
Assignments	10%	10
Midterm Exam	30%	30
Final Exam	50%	50
TOTAL	100%	100

Recommended Book:

- Aho, A. *et al.* (2015) **Compilers: Principles, techniques, and Tools**. India: Pearson India Education Services.

Reference Books:

- Holub, A.I. (2006) **Compiler design in C**. New Delhi: Prentice-Hall of India.
- Keith D. Cooper , Linda Torczon (2022) **Engineering a Compiler**. Elsevier Science

Benchmark:

- <https://web.stanford.edu/class/cs143/syllabus.html>
- <https://www.studocu.com/row/document/comsats-university-islamabad/compiler-construction/cs606-handouts-updated/9614386>



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LECTURE PLAN

Course Title: Compiler Construction

Course Code: CS-310T

SNo.	Week	Topics	Required Reading	Key Date
1	1st Week	Introduction to compilers ,Translators ,features of compiler, History of Compilers, Analysis of source program, Language processing system, Types of Language processing system, Types of Languages and compilers	AA -Chap 1: pg. 01 – 10 AH-Chap 1: Pg.16	
2	2 nd week	Compiler Phases with Diagrams, code Optimization and code generation Accessories of compiler.	AA -Chap 1: pg. 10 – 16 AH- chap9: pg-513, 585	
3	3 rd Week	Terminals, NonTerminals. Generate the Production Rules and also form the parse trees, Grammar for Identifiers, Variable Declaration.	AH-Chap 2: pg 52 & 53	Practice Problems
4	4 th Week	Grammar for Multiple data type declaration basic arithmetic operations, Loops If else Statements	AH-Chap 2: pg 54 AA-Chap 2: Pg.26,27,32	Assignment#1
5	5 th Week	Role, Features, Lexeme, Pattern,Token, Considerations for a simpler design of LA, Input Buffering Technique ways of implementation of LA	AA-Chap 3: Pg. 84-88	
6	6 th Week	Specification and recognition of Tokens ,Implementation of Tokens Implementation of Tokens Examples	AA-Chap 3: Pg 88-94	Quiz #1
7	7 th Week	Natural Language Grammar Formal Grammar, Context-free Grammar Processing of Grammars Example ofLeft mostand Right most derivation. Example of left and right most parse trees.	AA-Chap 3: Pg105,106, 168-170 AH-Chap2: Pg52-54	



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Week No.	Week Dates	Topics	Required Reading	Key Date
9	Midterm Examination (04-12-2023 to 09-12-2023)			
10	10 th Week	Left recursion , left factoring Simple and fast implementation of predictive parsers ,Non recursive Predictive Parsing	AA-Chap 4: Pg 176-187	Assignment #2
11	11 th Week	Finding the First and Follow sets Mapping of parsing table with Stack implementation. Predictive parser with examples LL(1) Grammar	AA-Chap 4: Pg 188- 190,192 AH-Chap 4: Pg 213-217	
12	12 th Week	Bottom up Parsing Technique Handles, handle pruning and stack implementation of SLR Shift-reduce parsers examples.	AA-Chap 4: Pg 195- 199,218-220 AH-Chap 5: Pg 338	Quiz #2 Assignment#3
13	13 th Week	Canonical collection parsing table and moves made by the LR parser Parsing table and LR parser examples	AA-Chap 4: Pg 220-225	
14	14 th Week	Over view of Syntax Directed Translation, Define Semantic Rules. Inheritance, synthesized attributes Inherited & synthesized attributes examples L& S attributed Grammar	AA-Chap 5: Pg 280-297	Quiz #3
15	15 th Week	Construction of Syntax Tree & Dependency graph, Translation Scheme. Over view of Type checking Type checking of statements Type checking of Expressions.	AA-Chap 5: Pg: 297-300 AA-Chap 6: Pg 300- 344	
16	16 th Week	Type checking of functions Example of statements and expression type checking Revision of Topics before Final Exam	AA-Chap 6: Pg 350-352	
Final Examination				

AA: Alfred V Aho ; **AH:** Allen Holub ; **KL:** Keith D. Cooper , Linda Torczon